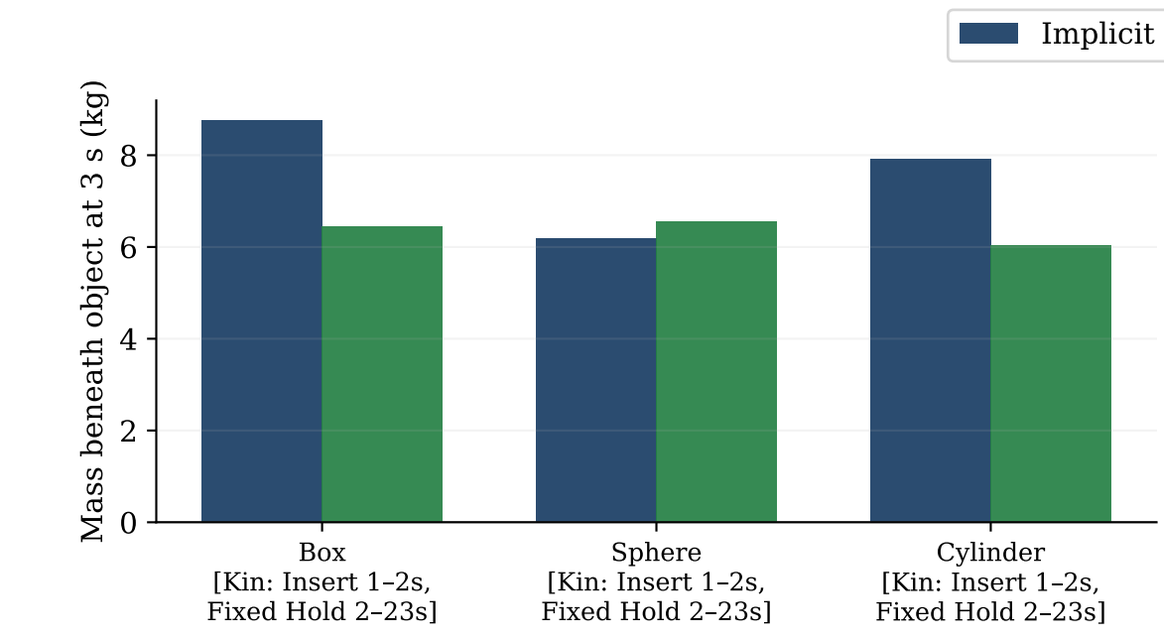
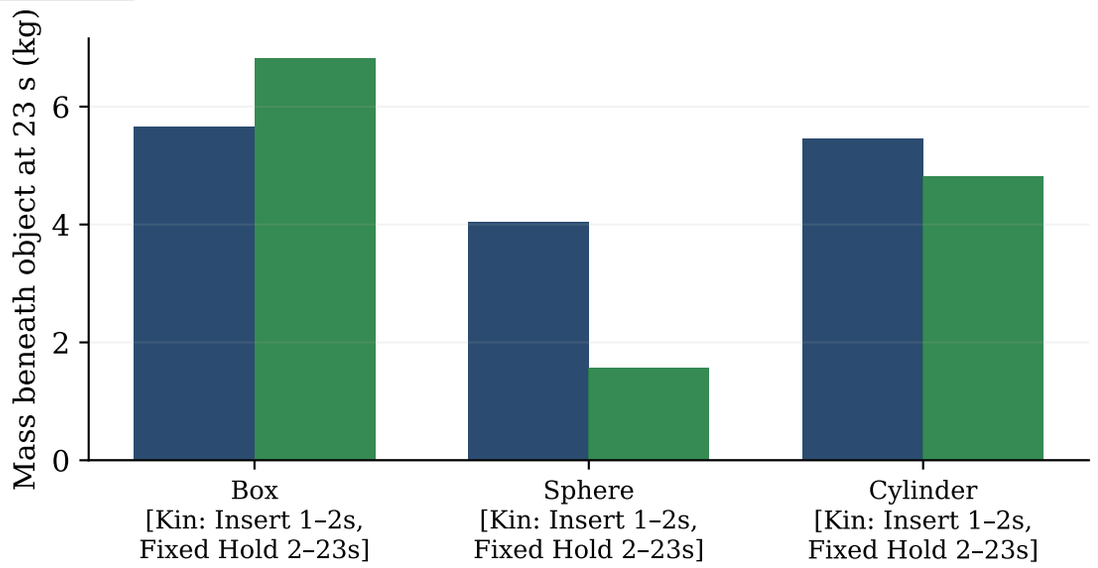
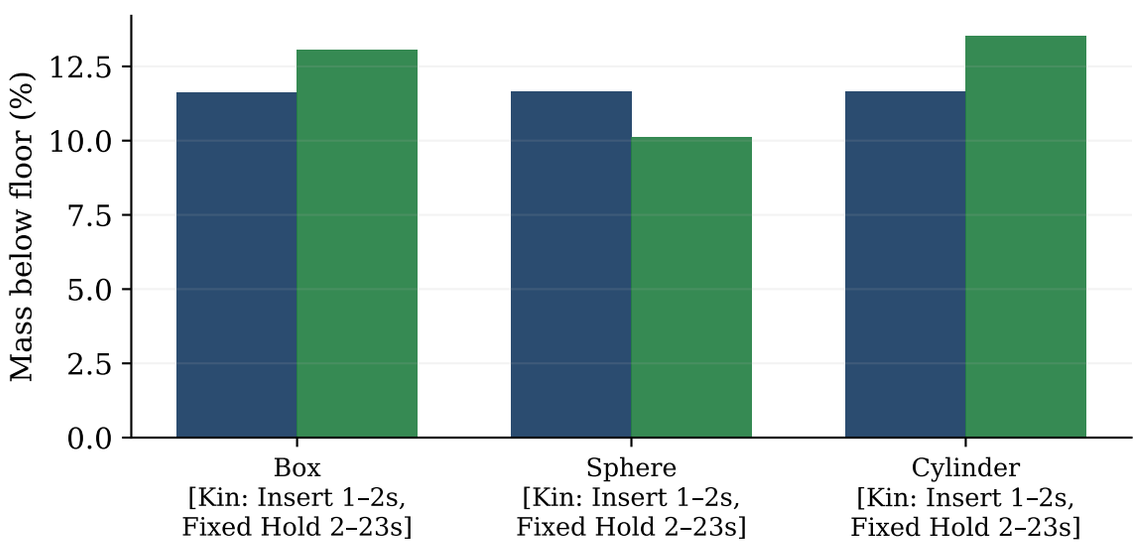
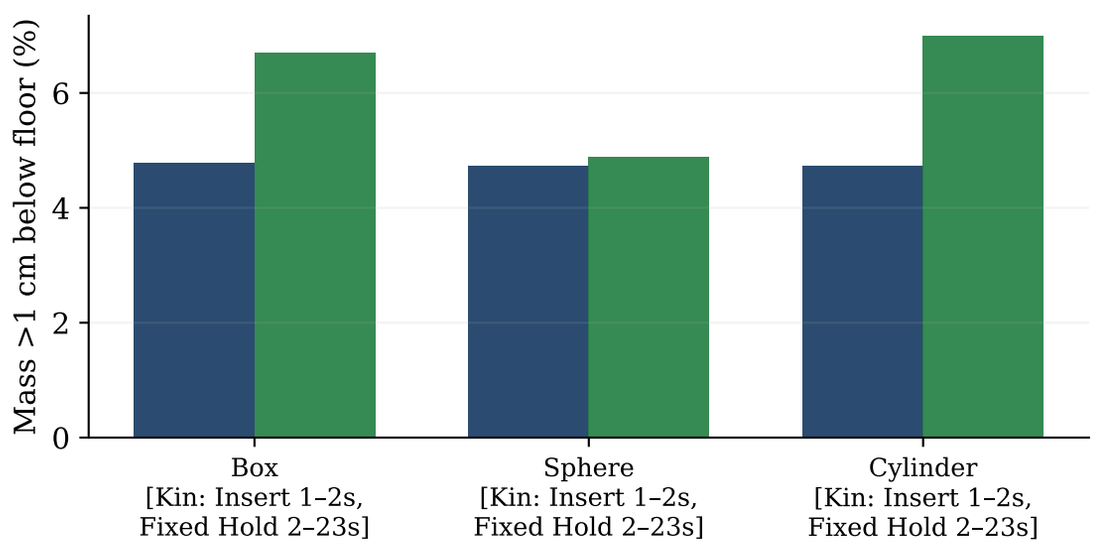
Kinematic Motion: Vertical Insertion (1–2 s) + Stationary Fixed-Pose Hold (2–23 s)









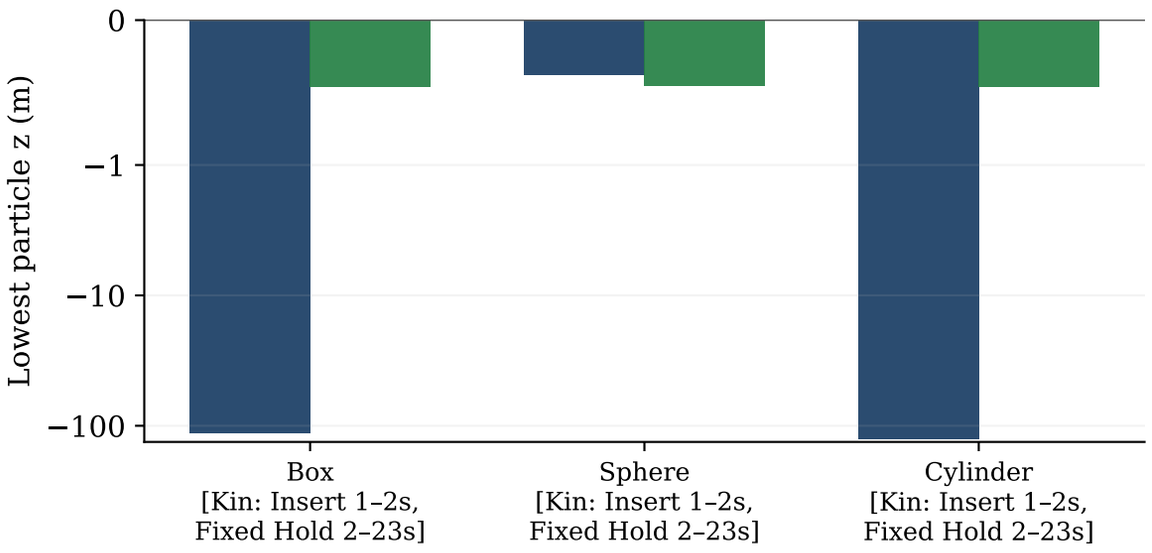


Figure 8: Frozen object-model checkpoint 10,800. Kinematic Motion: 2–23 s Fixed-Pose Hold for Box, Sphere, and Cylinder. Top row: physical mass beneath the object at 3 and 23 s. Middle row: mass below the z=−0.30 m floor. Last panel: lowest particle z at 23 s.